

L5

CLASS X - SCIENCE

LIFE PROCESSES

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LOW

MODERATE

HIGH





TODAY'S TARGET

- *Human Respiratory System*
- *Human Circulatory System*
- *Exchange of gases in plants*



LET'S TEST YOUR MEMORY



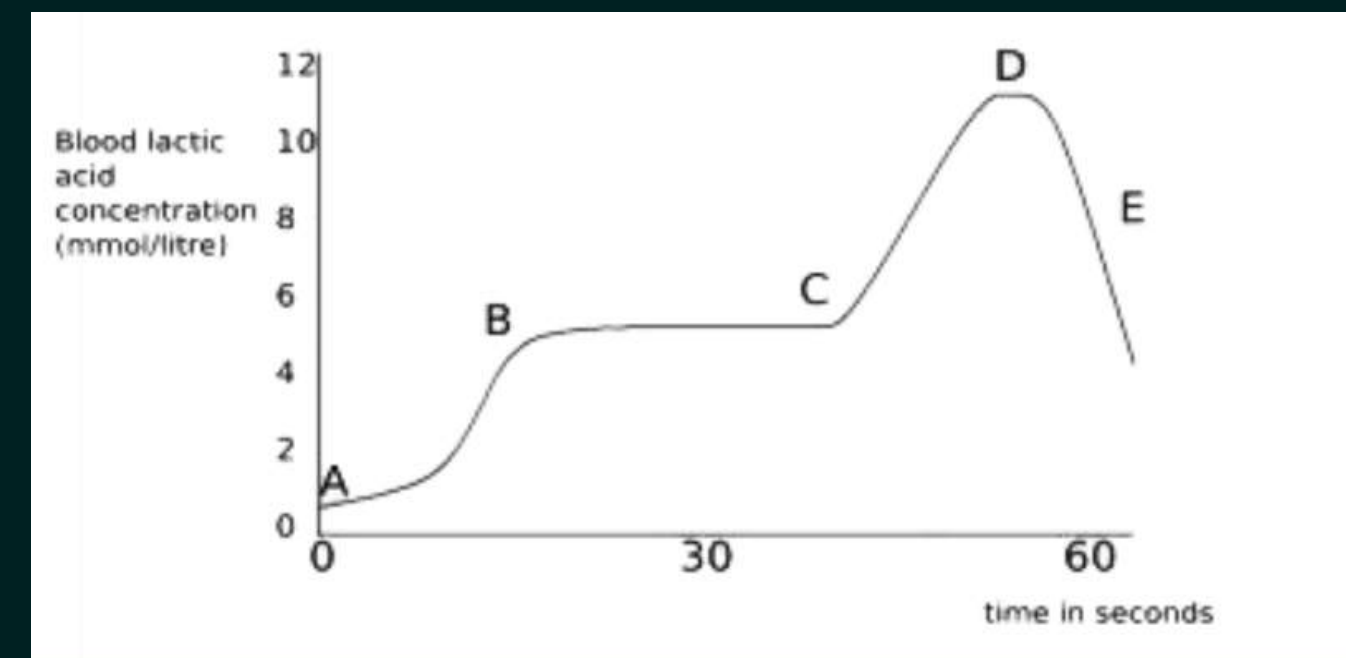
Q. All living cells require energy for various activities. This energy is available by the breakdown of simple carbohydrates either using oxygen or without using oxygen.

(i) Energy in the case of higher plants and animals is obtained by:

- (a) Breathing**
- (b) Tissue respiration**
- (c) Organ respiration**
- (d) Digestion of food**

(ii) The graph below represents the blood lactic acid concentration of an athlete during a race of 400 m and shows a peak at point D. Lactic acid production has occurred in the athlete while running in the 400 m race. Which of the following processes explains this event?

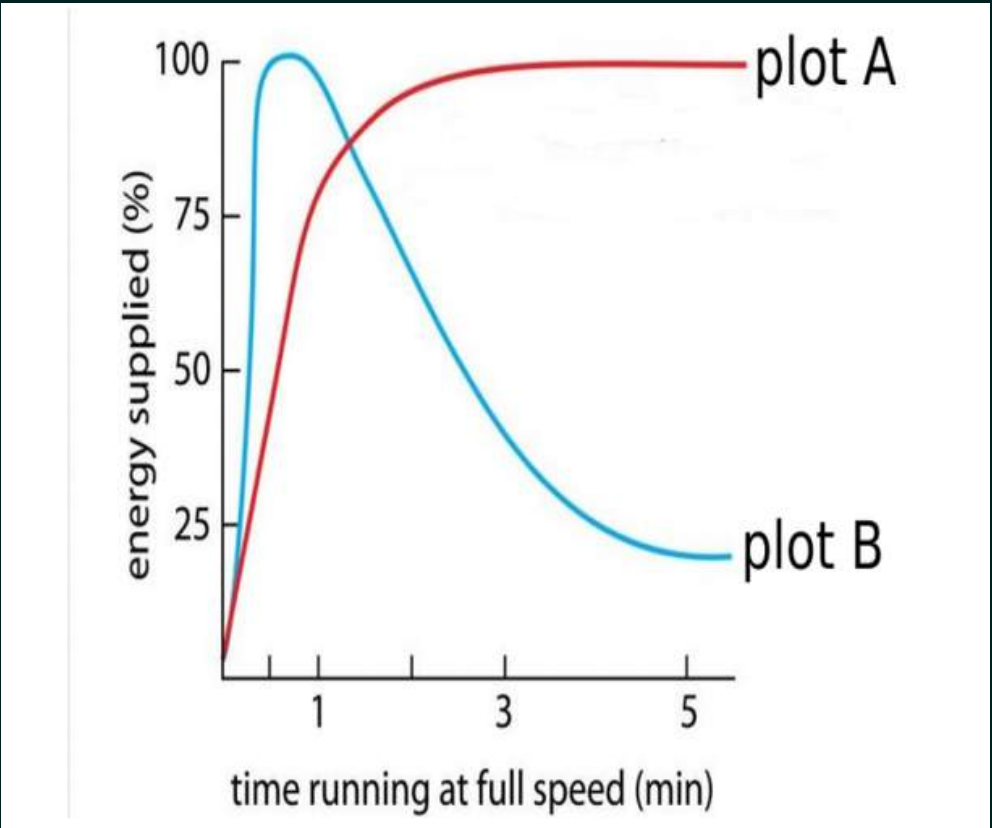
- (a) Aerobic respiration**
- (b) Anaerobic respiration**
- (c) Fermentation**
- (d) Breathing**



(iii) Study the graph below that represents the amount of energy supplied with respect to time while an athlete is running at full speed.

Choose the correct combination of plots and justification provided in the following table.

Option	Plot A	Plot B	Justification
(a)	Aerobic	Anaerobic	Amount of energy is low and inconsistent in aerobic and high in anaerobic
(b)	Aerobic	Anaerobic	Amount of energy is high and consistent in aerobic and low in anaerobic
(c)	Anaerobic	Aerobic	Amount of energy is high and consistent in aerobic and low in anaerobic
(d)	Anaerobic	Aerobic	Amount of energy is high and inconsistent in anaerobic and low in aerobic

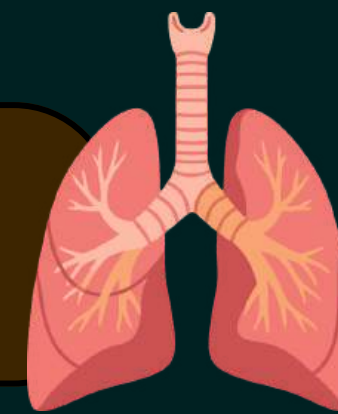


(iv) The characteristic processes observed in anaerobic respiration are:

- (i) Presence of oxygen
- (ii) Release of carbon dioxide
- (iii) Release of energy
- (iv) Release of lactic acid

- (a) (i), (ii) only
- (b) (i), (ii), (iii) only
- (c) (ii), (iii), (iv) only
- (d) (iv) only

Breathing vs Respiration

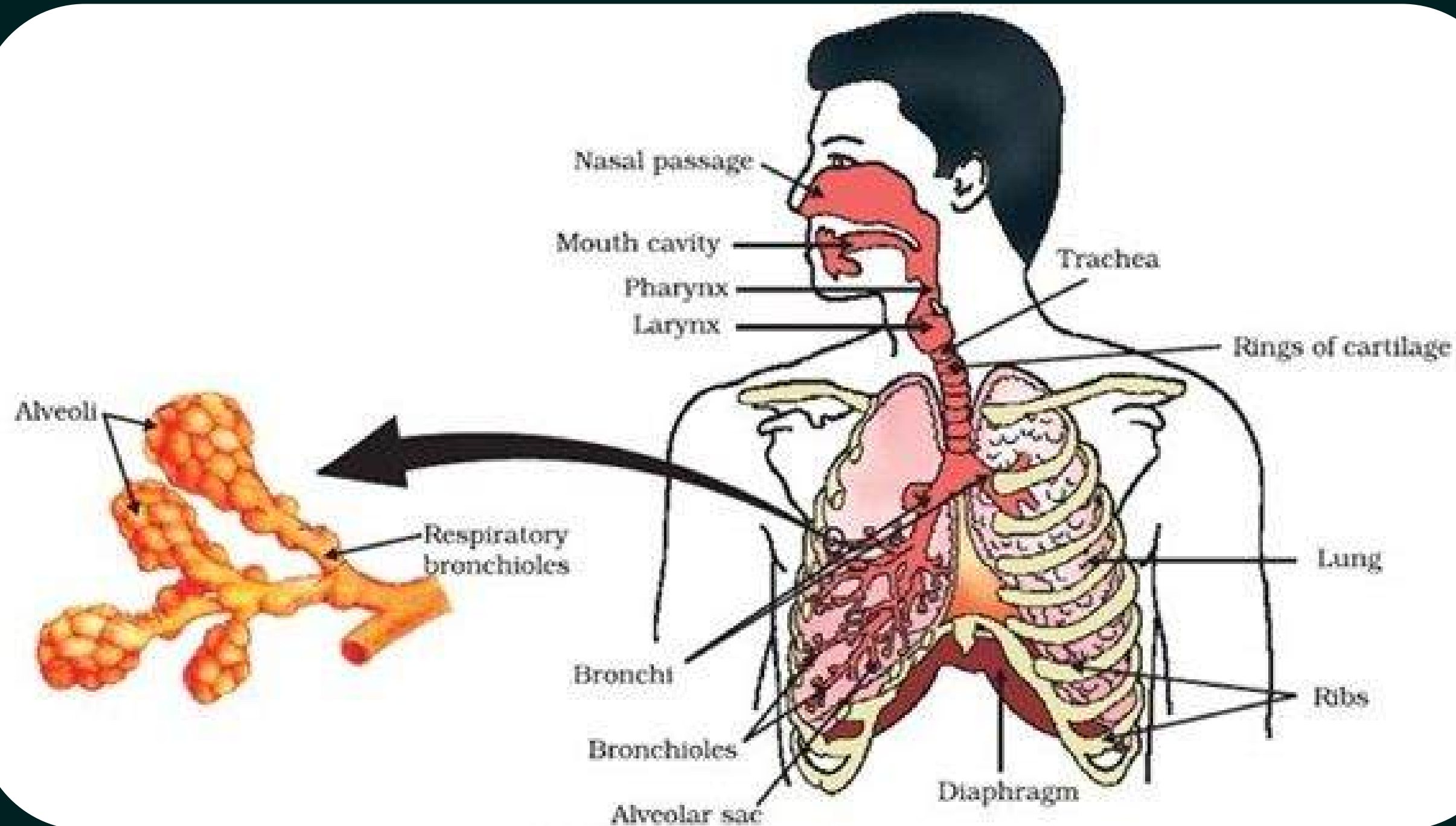


Breathing	Respiration
Breathing is the process by which oxygen is taken into the body and carbon dioxide is released out.	Respiration is the process in which food is broken down inside the cells to release energy.
It is a physical process.	It is a biochemical process.
It occurs in respiratory organs like lungs.	It occurs inside the cells.
It involves exchange of gases.	It involves breakdown of glucose.
Energy is not released during breathing.	Energy is released during respiration.
Example: Inhalation and exhalation.	Example: Breakdown of glucose to release energy.

Human Respiratory System

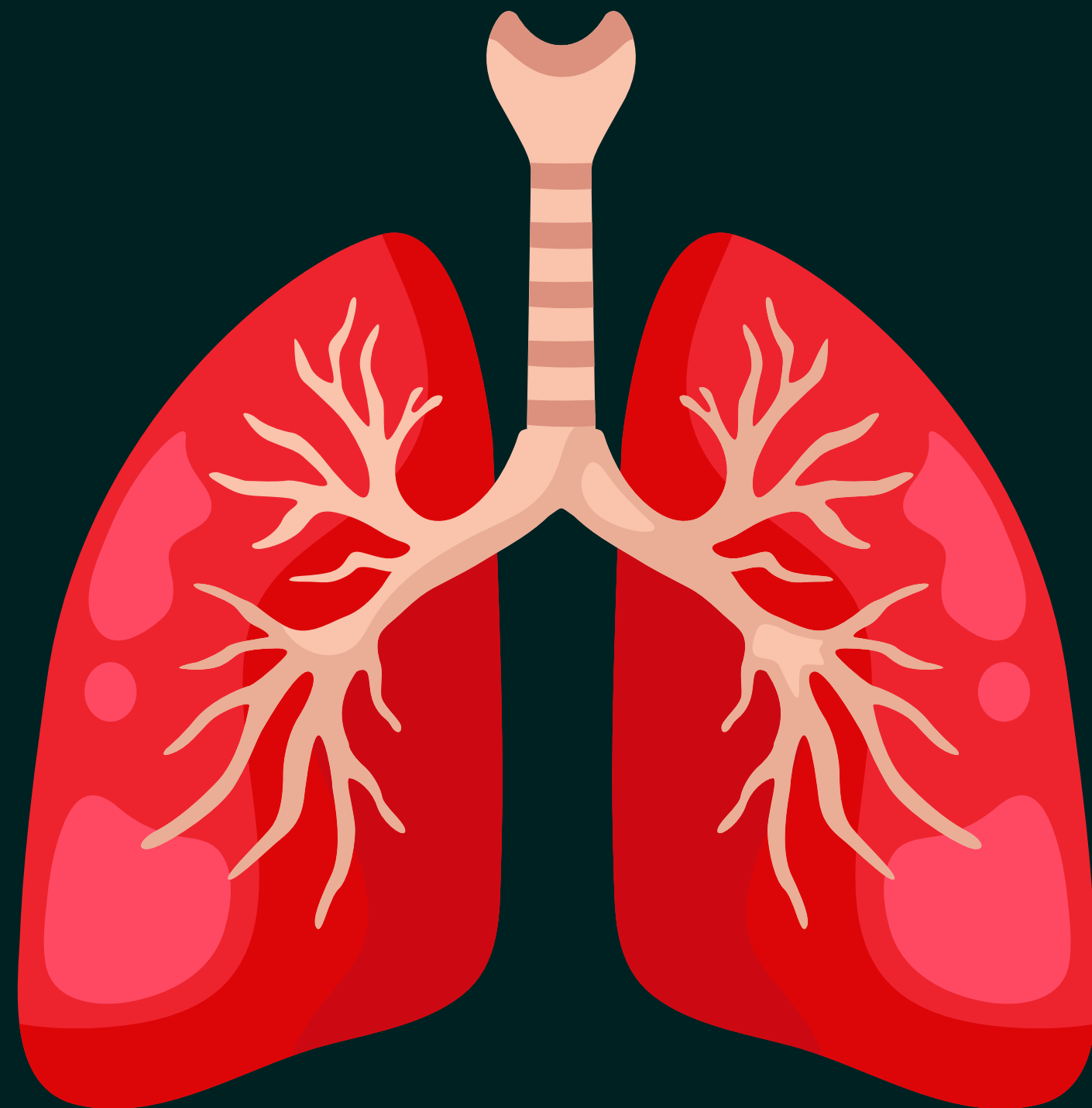


Human Respiratory System

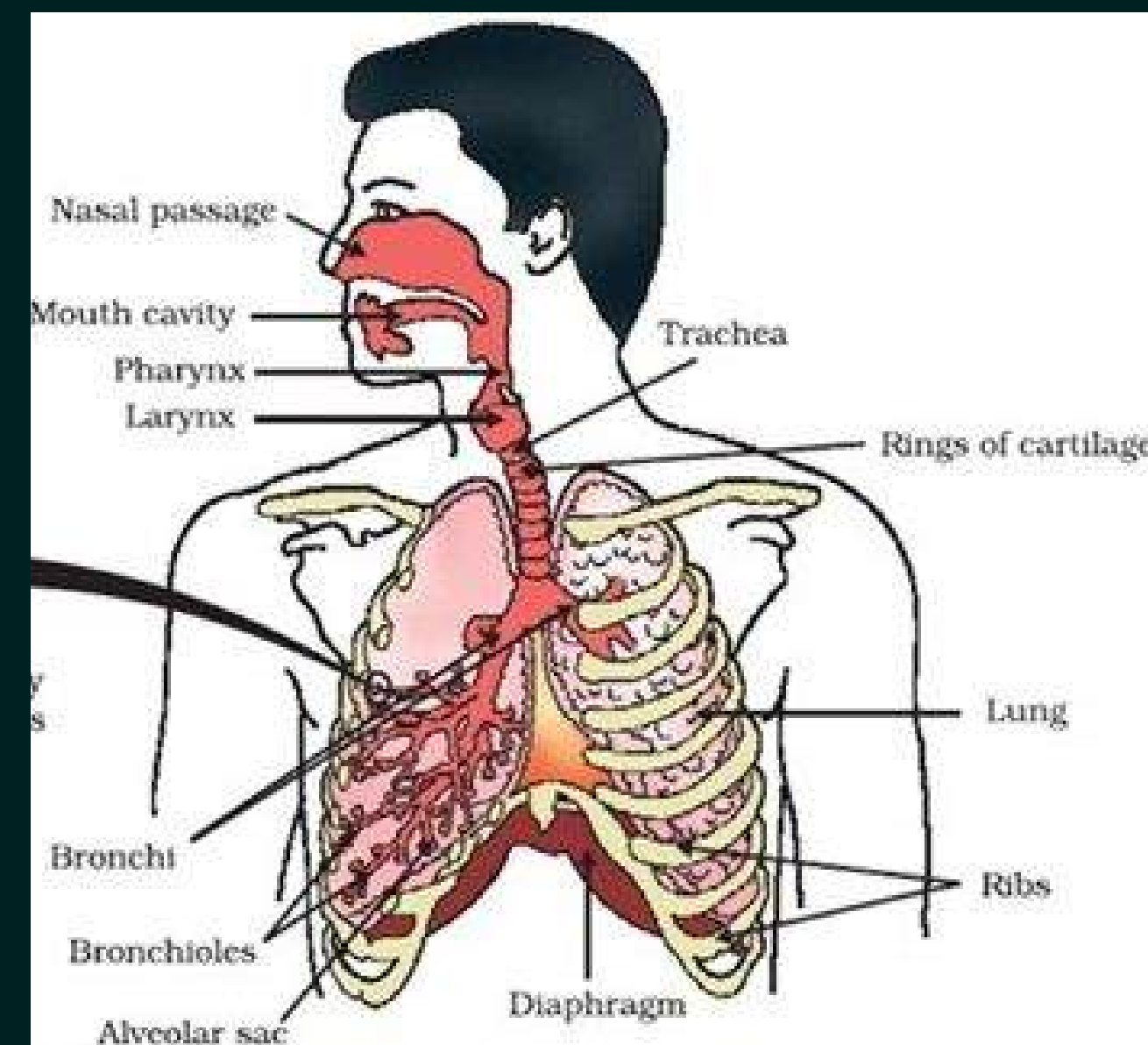
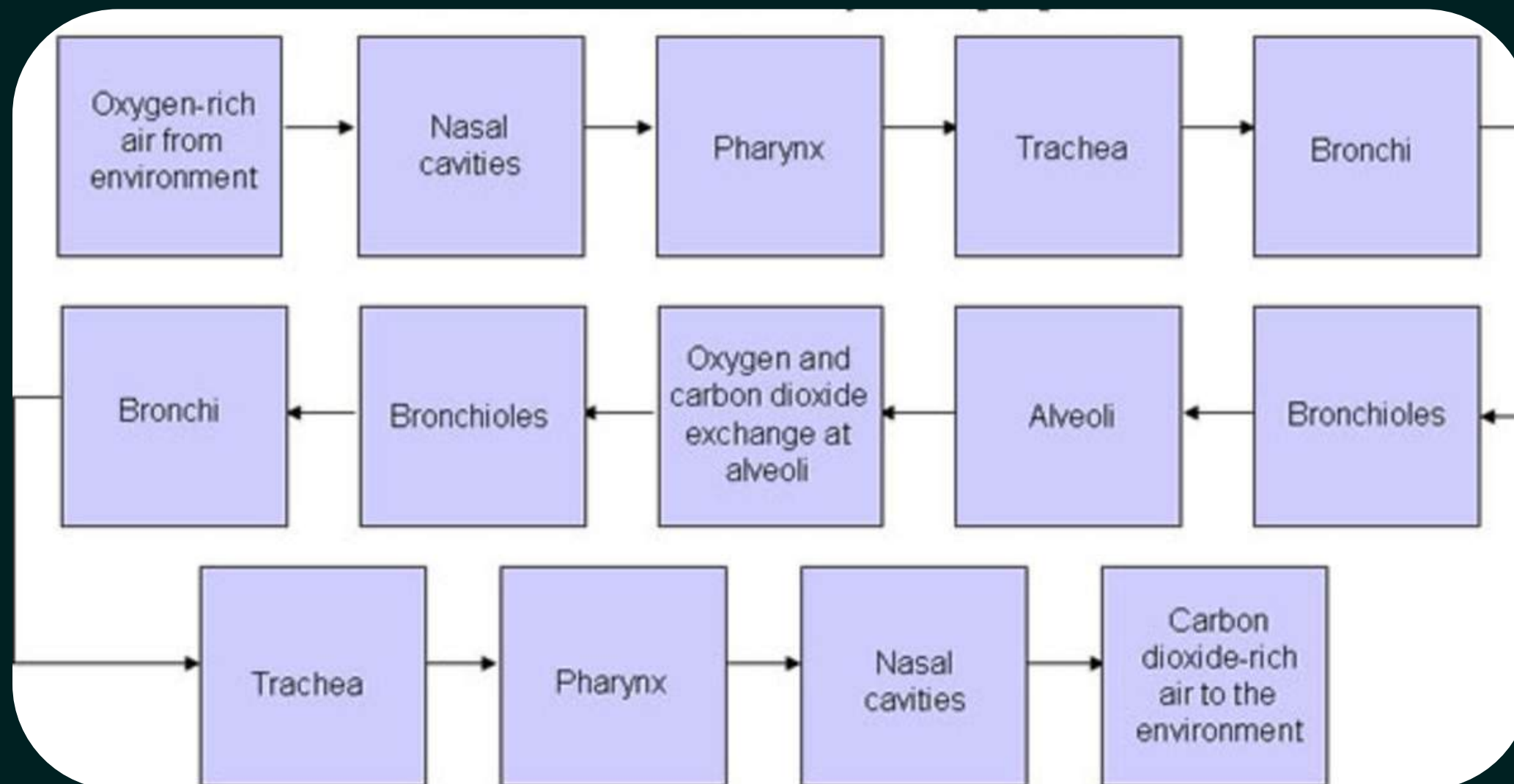


Human Respiratory System

- The human respiratory system is the biological system responsible for taking in oxygen and removing carbon dioxide from the body.
- It helps us breathe, and it's essential for cellular respiration, which produces the energy our body needs to function.



Exchange of gases in Human Body



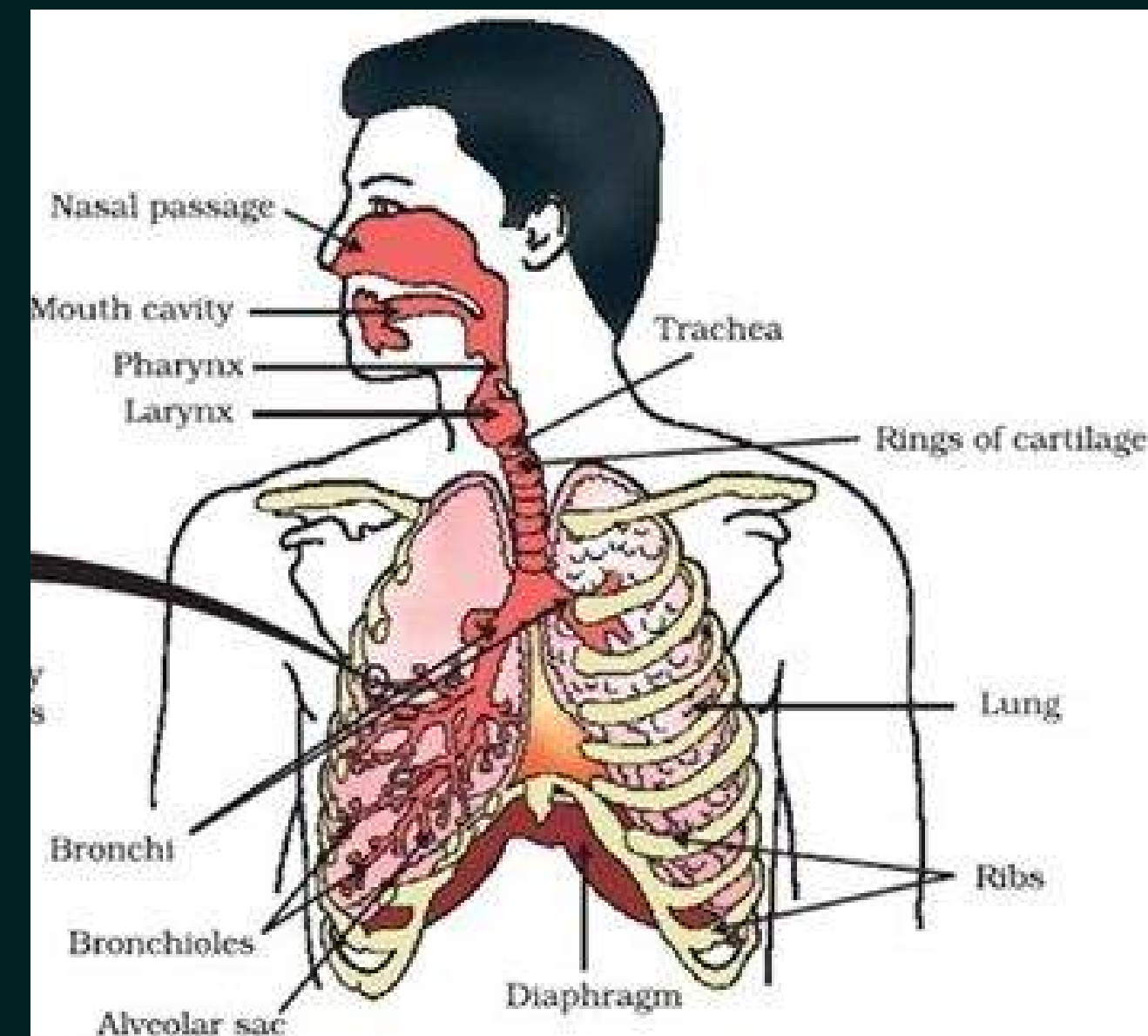
Passage of air through Respiratory System:

Nostril: Air enters the body through the nostrils.

Nasal Passage: The air passing through the nostrils is filtered by fine hairs that line the passage. The passage is also lined with mucus, which helps in trapping dust and other particles.

Pharynx: Pharynx is the common passage for food and air. Air from the nasal passage passes through the pharynx and moves towards the windpipe.

Larynx: Larynx is also called the voice box. It connects the pharynx to the trachea and helps in producing sound.



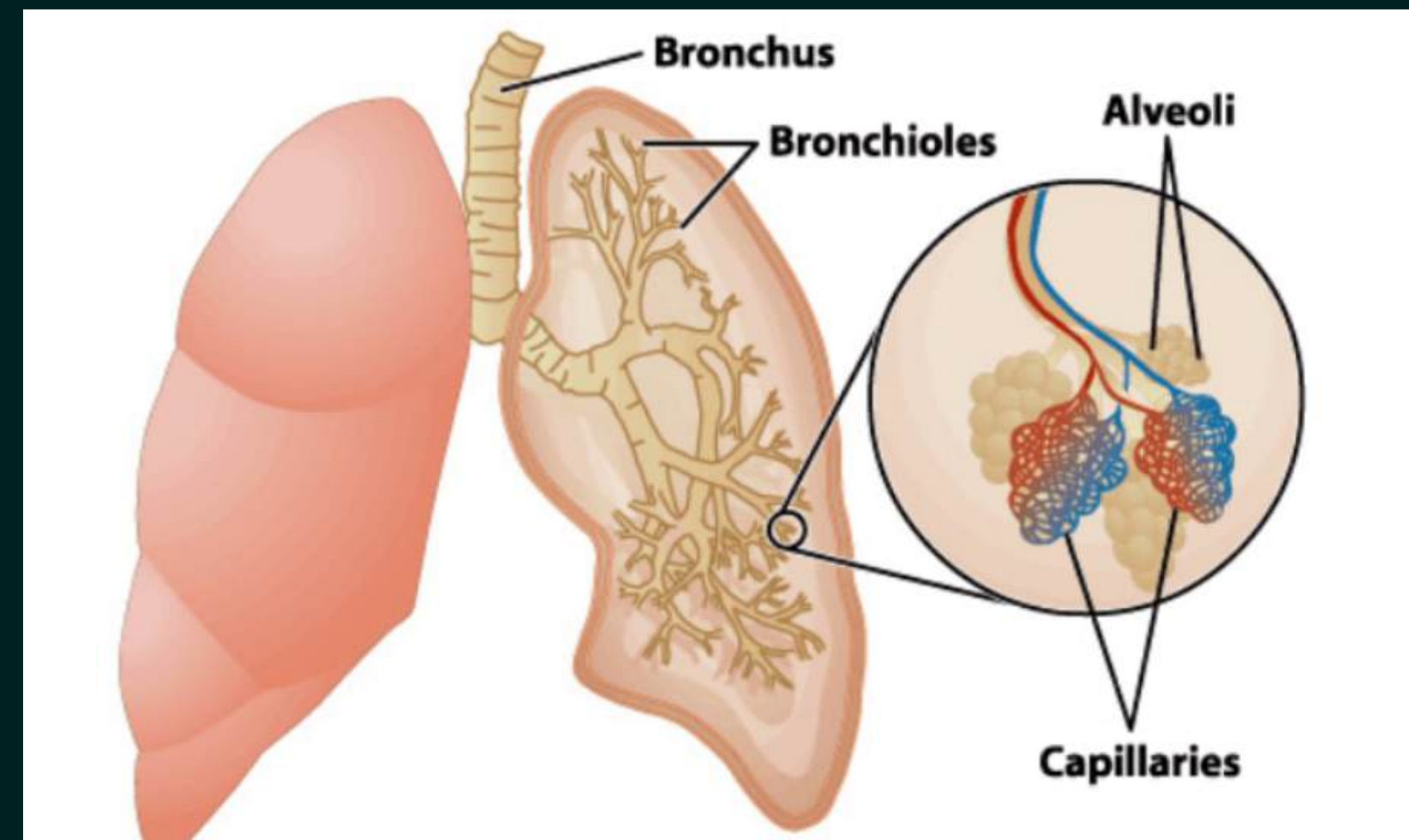
Trachea / Air Passage: Trachea is the windpipe through which air moves towards the lungs. Rings of cartilage are present in the trachea/air passage.

These rings prevent the air passage from collapsing.

Bronchi: The trachea divides into two bronchi, one entering each lung.

Bronchioles: Inside the lungs, the bronchi divide into smaller and smaller tubes called bronchioles

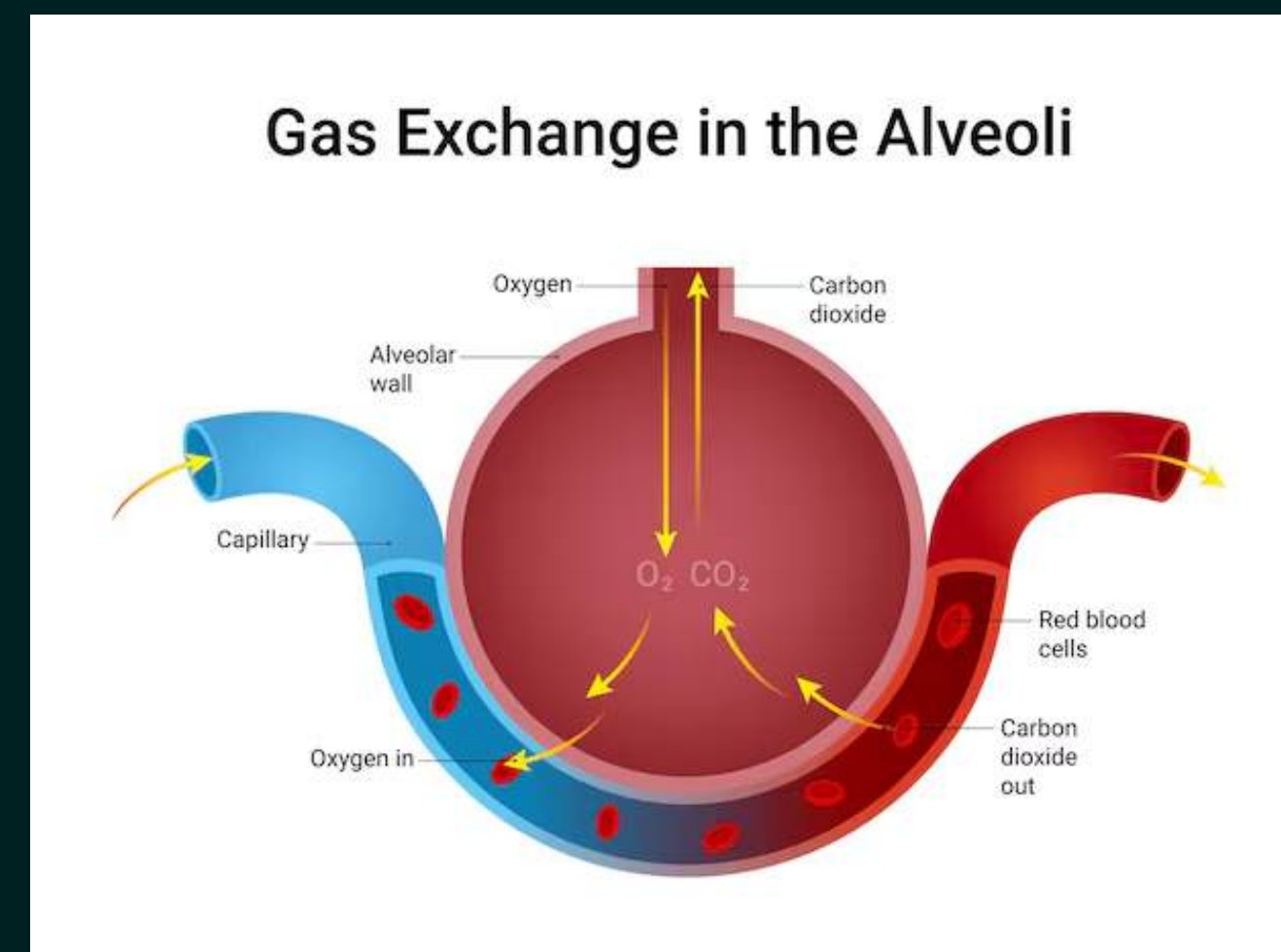
Alveoli: Bronchioles finally end in balloon-like structures called alveoli. Alveoli provide a surface where exchange of gases takes place.



They are the site of gas exchange in the lungs

Alveolus Gas Exchange

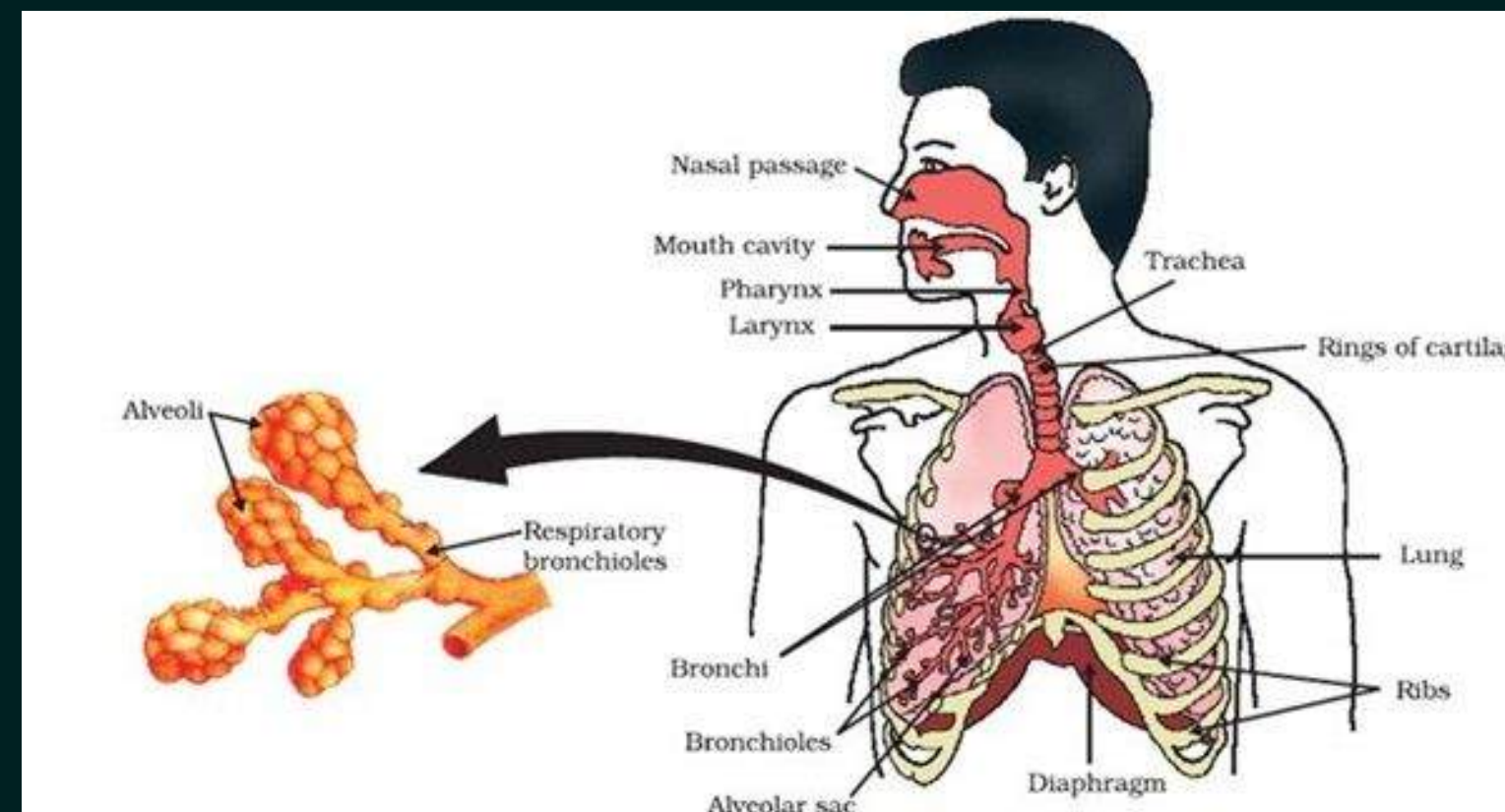
The blood brings carbon dioxide from the rest of the body for release into the alveoli, and the oxygen in the alveolar air is taken up by blood in the vessels to be transported to all the cells in the body.



Alveoli are suitable for gas exchange because:

1. They provide a large surface area.
2. They have thin walls.
3. They are surrounded by blood capillaries.
4. They have moist surfaces for diffusion of gases.

Blood Vessels Around Alveoli: The walls of alveoli contain an extensive network of blood vessels. Oxygen from the alveolar air is taken up by blood. Carbon dioxide from the blood is released into the alveoli.



Haemoglobin

- A respiratory pigment
- Iron containing protein
- Gives red color to red blood cells.

Affinity of Haemoglobin for various gases: $\text{CO} > \text{O}_2 > \text{CO}_2$

Red Blood Cells

- Haemoglobin present inside RBCs binds with oxygen.
- It takes up oxygen from the lungs.
- RBCs transport oxygen to all parts of the body.
- After reaching body tissues, haemoglobin releases oxygen for cellular respiration.
- Carbon dioxide is mostly transported in dissolved form in the blood because it is more soluble in water than oxygen.



Fact: The biconcave shape of RBCs increases surface area for efficient gas exchange!

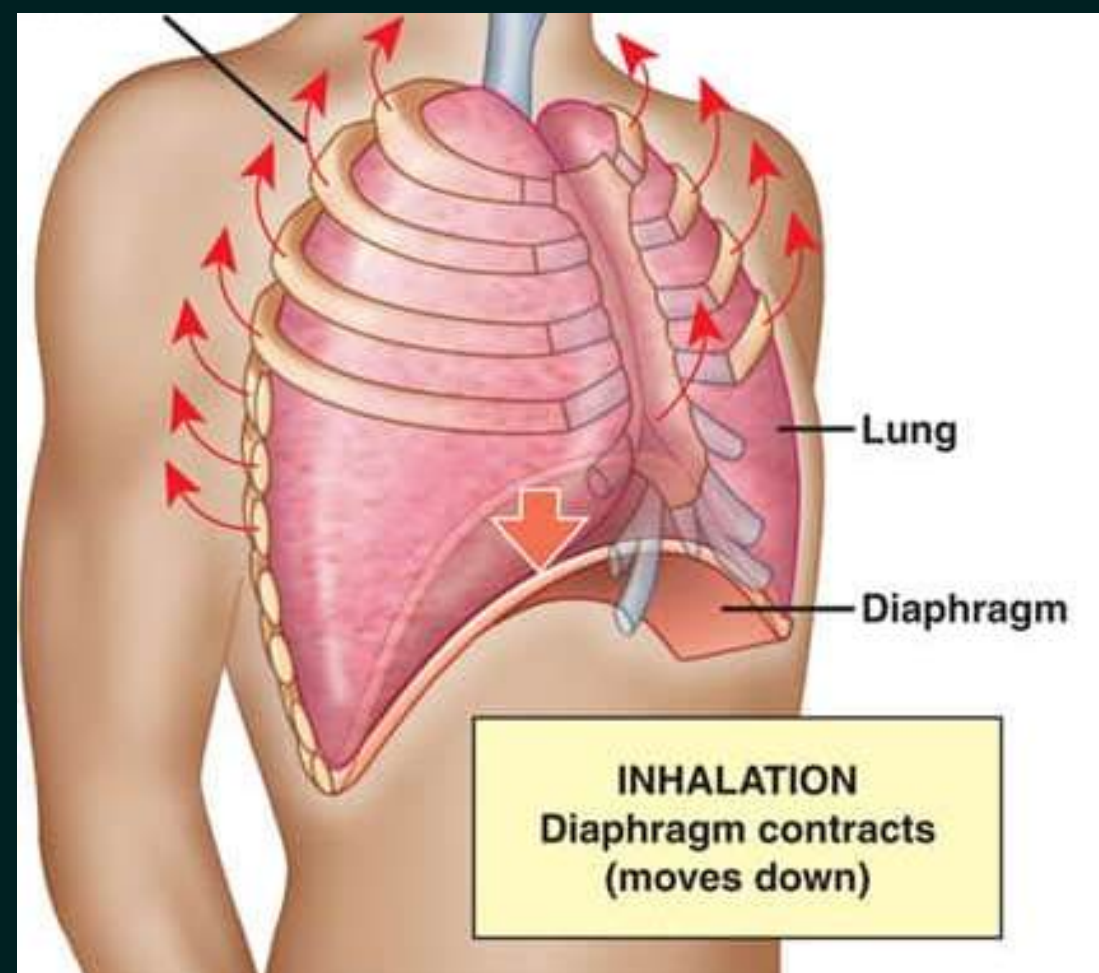
Soch ke bataao?



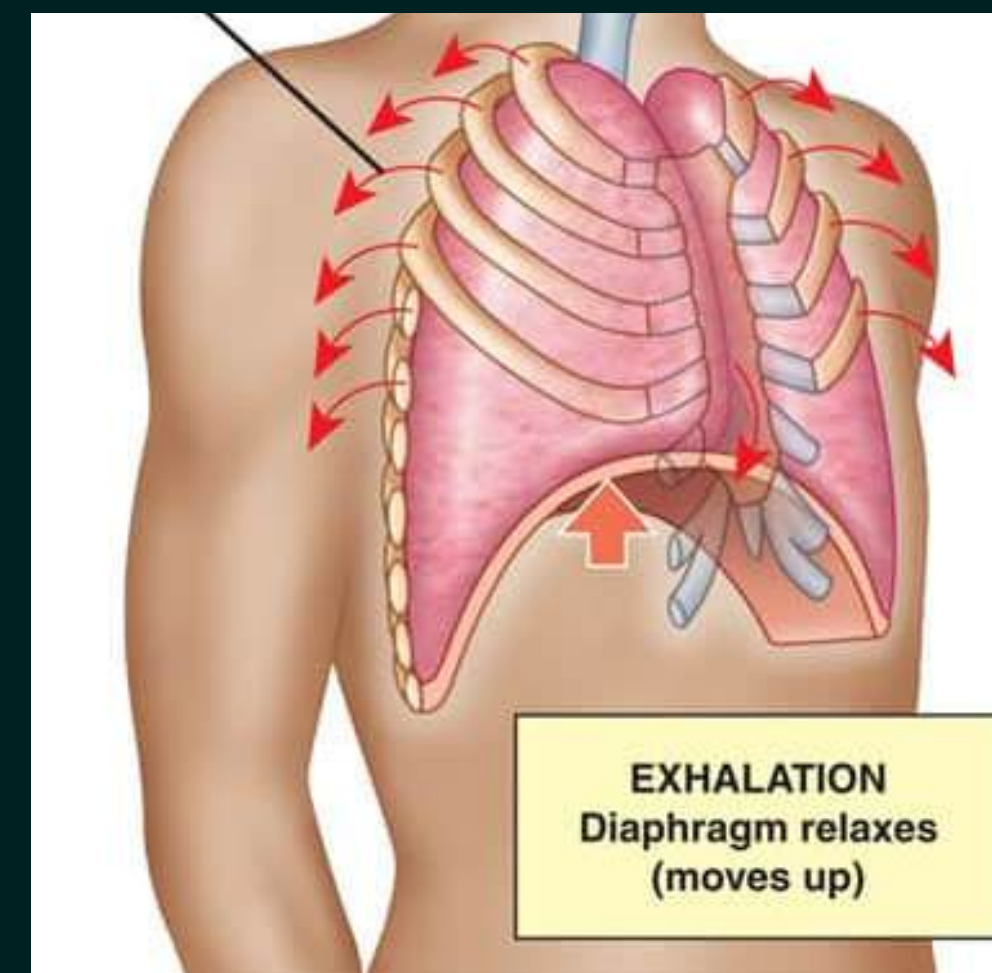
Lack of hemoglobin will lead to weakness?
YES or NO??

Breathing Mechanism (Inhalation and Exhalation)

DIAPHRAGM: A skeletal muscle that separates the chest from the abdomen and helps in breathing.



Inhalation: Ribs move up, diaphragm flattens
Chest cavity increases → air is sucked into lungs



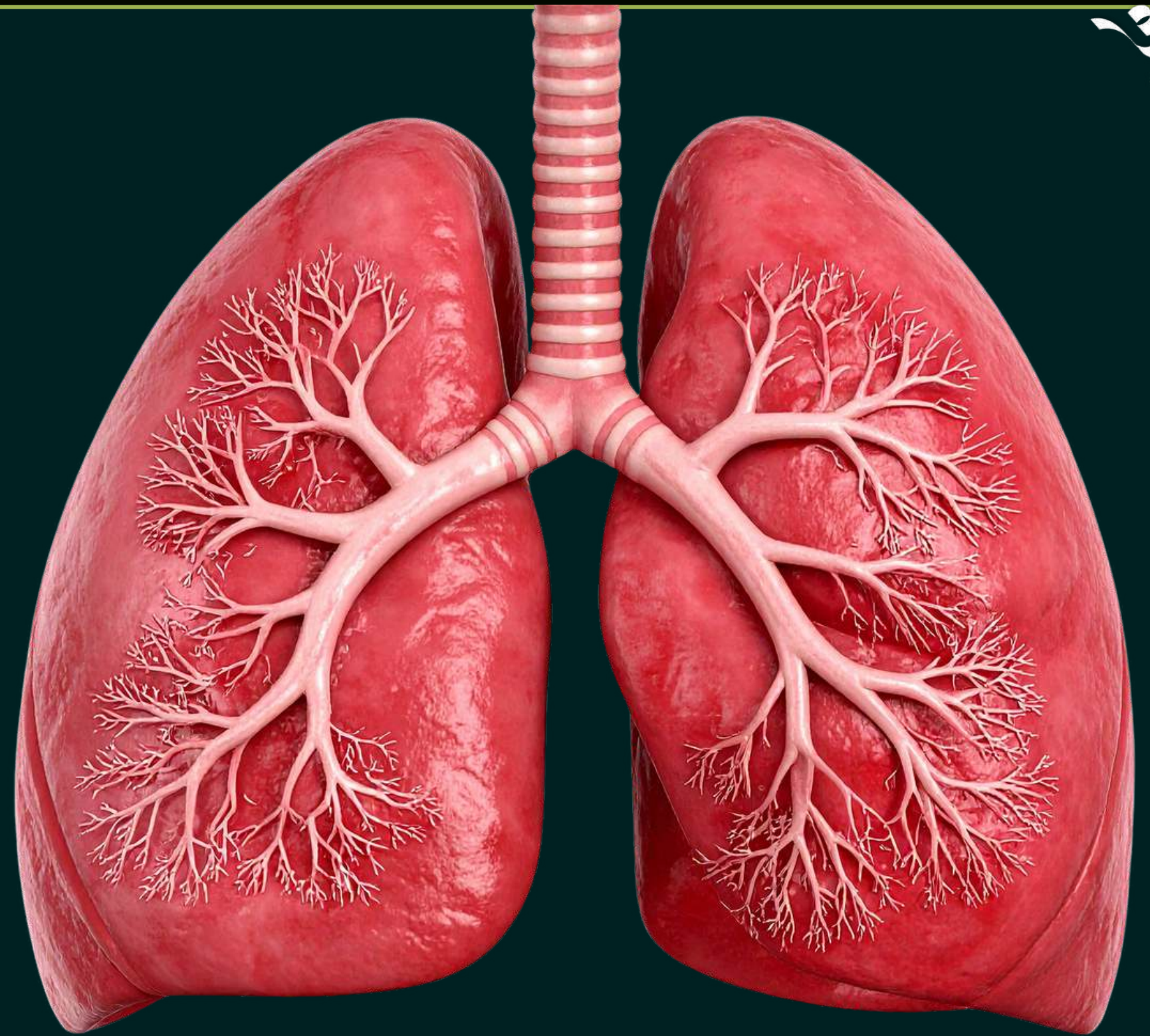
Exhalation: Ribs move down, diaphragm relaxes
(domes up)
Chest cavity decreases → air is pushed out

Inhalation V/S Exhalation

Inhalation	Exhalation
It is the process of breathing in air.	It is the process of breathing out air.
Ribs move upwards and outwards.	Ribs move downwards and inwards.
Diaphragm contracts and flattens.	Diaphragm relaxes and becomes dome-shaped/arches upwards.
Volume of the chest cavity increases.	Volume of the chest cavity decreases.
Air pressure inside the chest cavity decreases.	Air pressure inside the chest cavity increases.
Air is sucked into the lungs.	Air is pushed out of the lungs.

Q. Why lungs have residual air

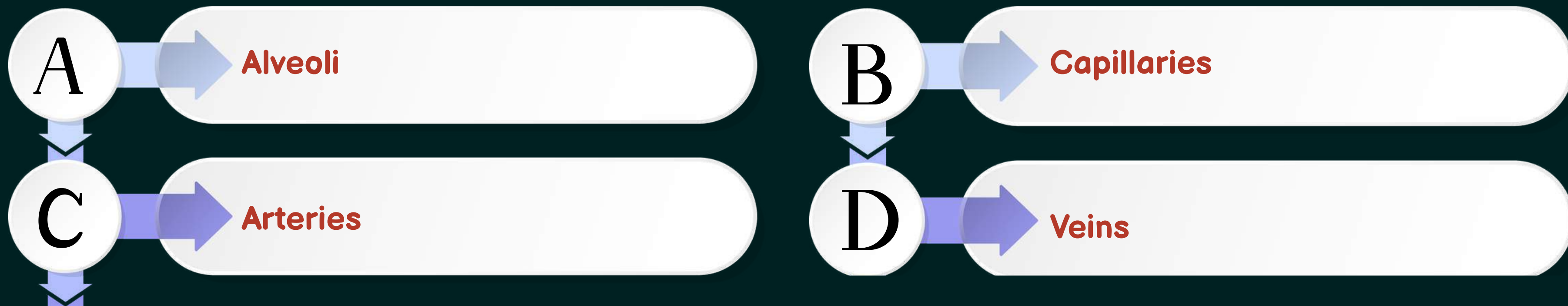
The lungs always contain some residual air so that there is enough time for oxygen to be absorbed into the blood and carbon dioxide to be released from the blood into the alveoli.



LET'S TEST YOUR MEMORY



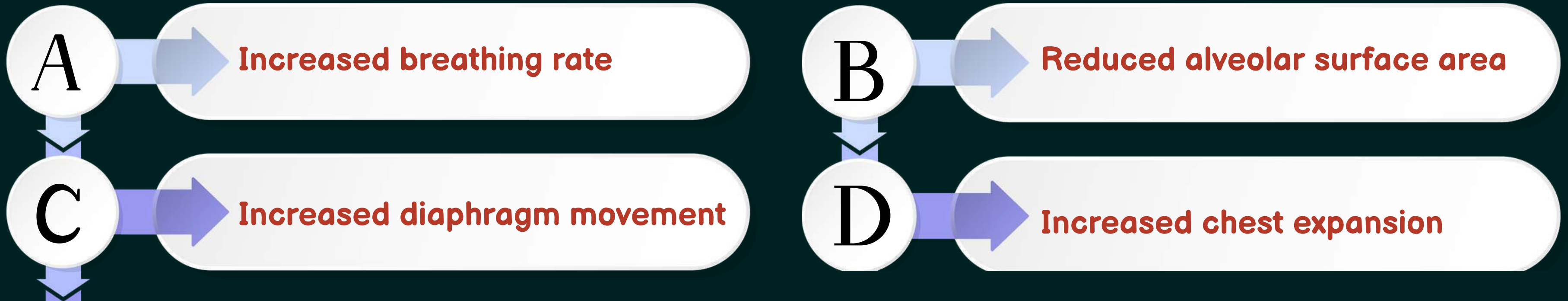
Q. One-cell thick blood vessels are known as:



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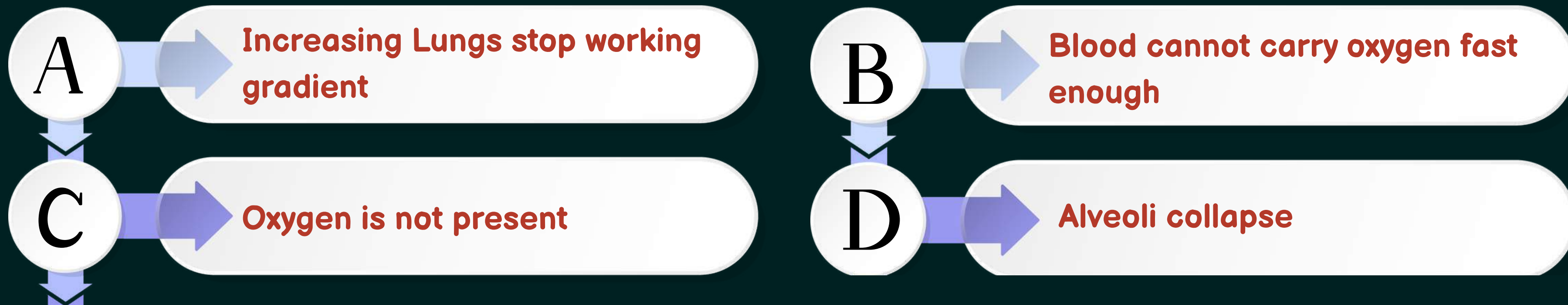
Q. A person has normal lung volume but reduced oxygen absorption. Which is the most likely cause?



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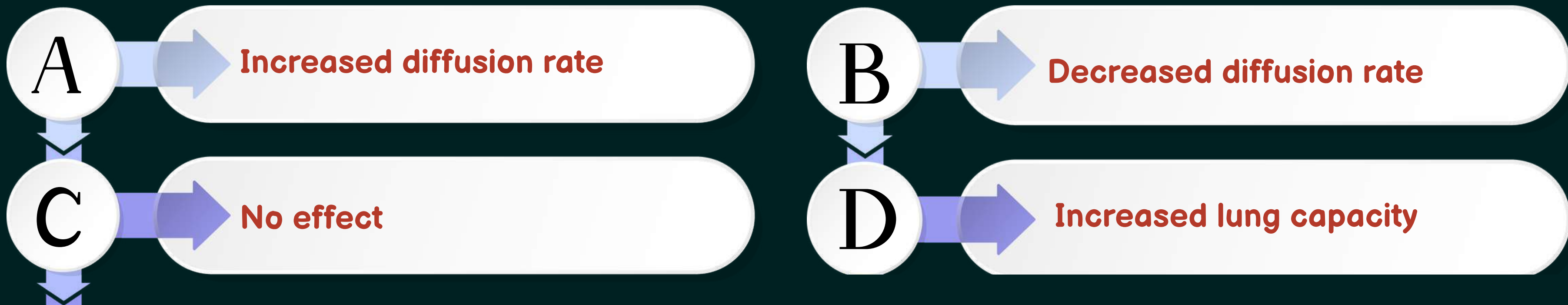
Q. During heavy exercise, breathing becomes rapid. However, oxygen delivery still becomes limiting. Why?



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Q. If alveolar walls become thicker, which immediate change occurs?



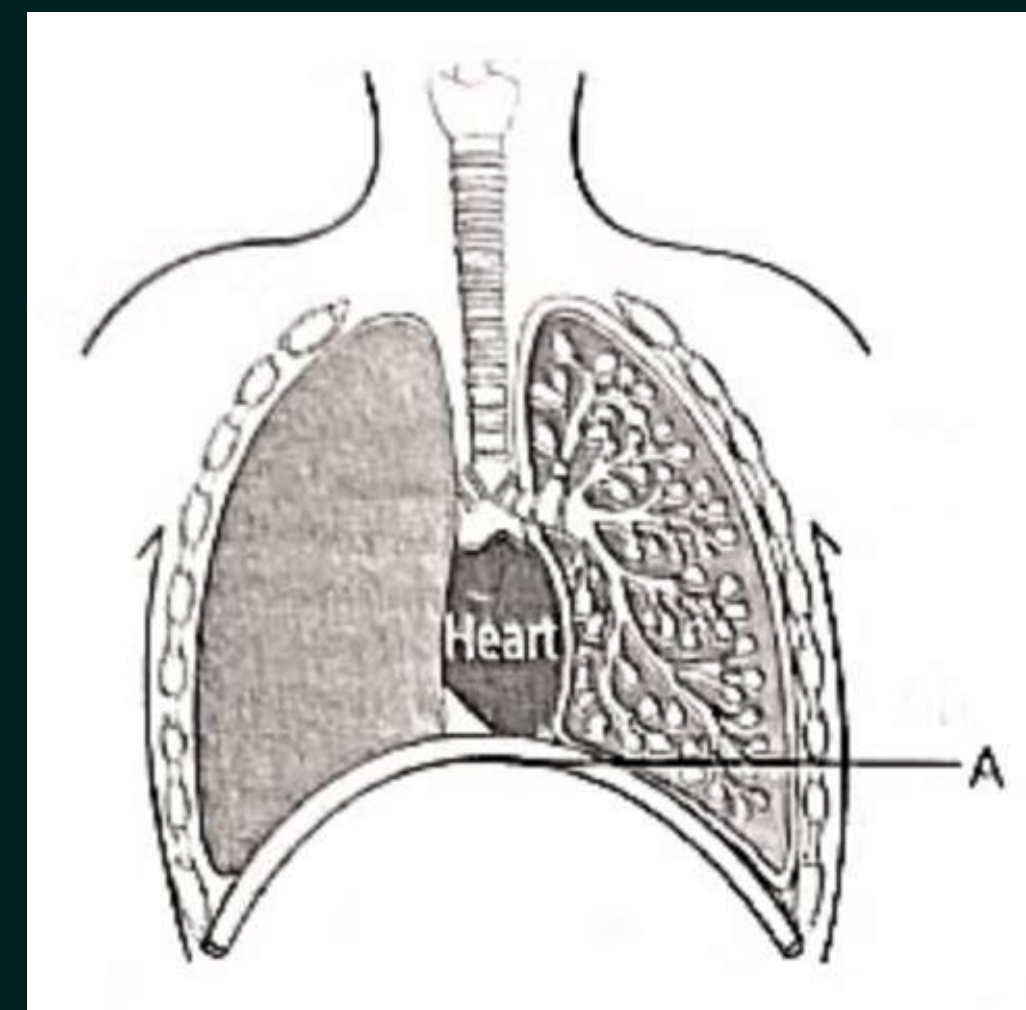
LET'S TEST YOUR MEMORY



Q. Which of the following statements are correct in reference to the role of A, shown in the given diagram, during a breathing cycle in human beings?

- (i) It helps to decrease the residual volume of air in lungs.
- (ii) It flattens as we inhale.
- (iii) It gets raised as we inhale.
- (iv) It helps the chest cavity to become larger.

- (a) (ii) and (iv)
- (b) (iii) and (iv)
- (c) (i) and (iii)
- (d) (i), (ii) and (iv)



LET'S TEST YOUR MEMORY



Q. Two individuals inhale the same volume of air:

- **Person A: Healthy lungs**
- **Person B: Reduced alveolar surface**

Q1. Who absorbs more oxygen? Why?

Q2. Why can't Person B compensate by breathing faster?

1. Because healthy lungs have a larger alveolar surface area, so more oxygen diffuses into the blood.

2. because the alveolar surface area is reduced. So, even if more air enters, less oxygen can pass into the blood.

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Q. Why is surface area more important than breathing rate for gas exchange?

Surface area is more important because the exchange of gases takes place through the surface of alveoli.

If alveolar surface area is large, more oxygen can diffuse into the blood at the same time. But if the surface area is reduced, then even faster breathing cannot help much because less area is available for gas exchange.

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Q. Why does breathing continue even when oxygen demand is low (like during sleep)?

Breathing continues even during sleep because body cells still need oxygen for cellular respiration and continue producing carbon dioxide. Even when oxygen demand is low, oxygen is needed to release energy from food, and carbon dioxide must be removed from the body.

LET'S TEST YOUR MEMORY



Q. Predict what happens if alveoli lose moisture.

If alveoli lose moisture, the diffusion of gases will become less efficient. Oxygen will not dissolve properly and pass into the blood easily. Similarly, carbon dioxide removal will also become slow.

LET'S TEST YOUR MEMORY



Q. Give reasons for the following:

- (a) Rings of cartilage are present in the throat.**
- (b) Lungs always contain a residual volume of air.**
- (c) The diaphragm flattens and ribs are lifted up when we breathe in.**
- (d) Walls of alveoli contain an extensive network of blood vessels.**

Ans. (a) To keep the windpipe open and prevent its collapse, ensuring continuous airflow to the lungs.

(b) To allow continuous exchange of gases and prevent lung collapse.

(c) To increase chest cavity volume, creating low pressure so air can enter the lungs.

(d) To allow efficient exchange of oxygen and carbon dioxide between lungs and blood.

LET'S TEST YOUR MEMORY



Q. Why is respiratory pigment needed in multicellular organisms with large body sizes?

Ans. In multicellular organisms with large body sizes, diffusion alone is not sufficient to supply oxygen to all body cells efficiently. Hence, a respiratory pigment like haemoglobin is required.

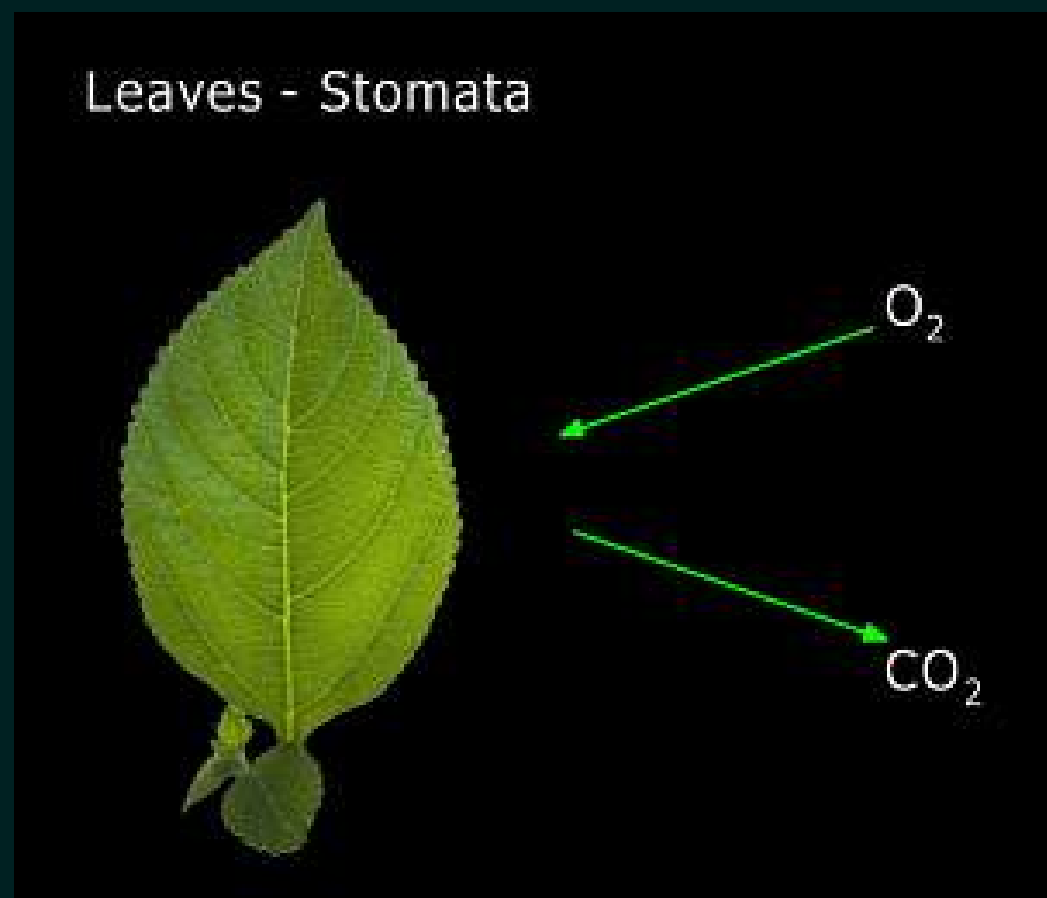
Haemoglobin, present in red blood cells (RBCs), has a high affinity for oxygen. It helps in absorbing oxygen from the lungs and transporting it to all body tissues. It also carries carbon dioxide from tissues back to the lungs for exhalation

Exchange of gases in Plants

Respiration in plants is simpler than the respiration in animals.

Gaseous exchange occur through :

(a) Stomata in leaves



(b) Lenticles in stems

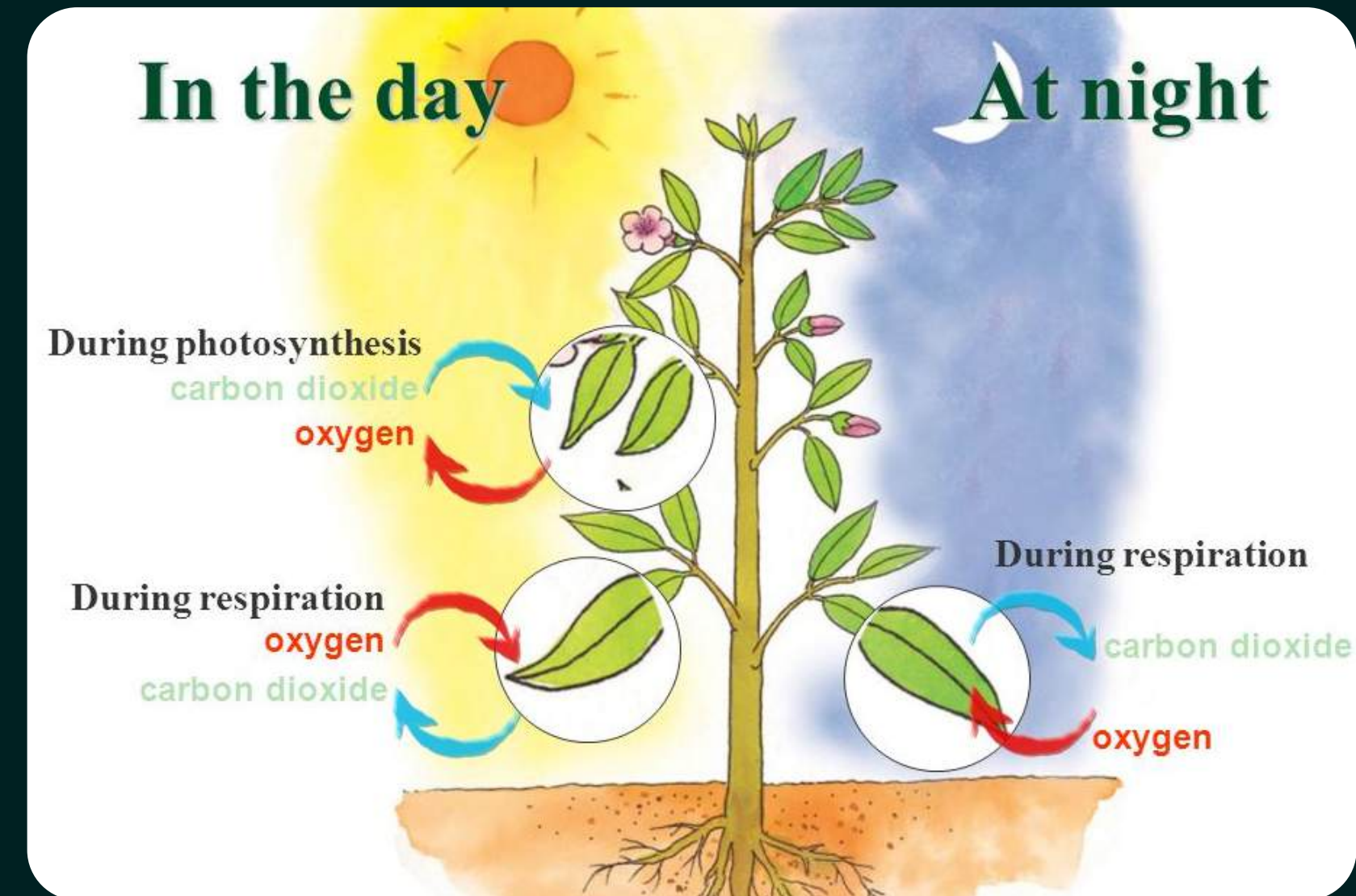


(c) General surface of Roots



Exchange of gases in Plants

Daytime ☀️	Night time 🌙
Photosynthesis + Respiration	Only Respiration
CO ₂ is taken in for photosynthesis	CO ₂ is released as a byproduct of respiration
O ₂ is released as a byproduct of photosynthesis	O ₂ is used for respiration
Oxygen released, CO ₂ absorbed	Carbon dioxide released, O ₂ absorbed



Roots take oxygen from the air spaces present between soil particles. Carbon dioxide produced during respiration diffuses out through the roots.

LET'S TEST YOUR MEMORY



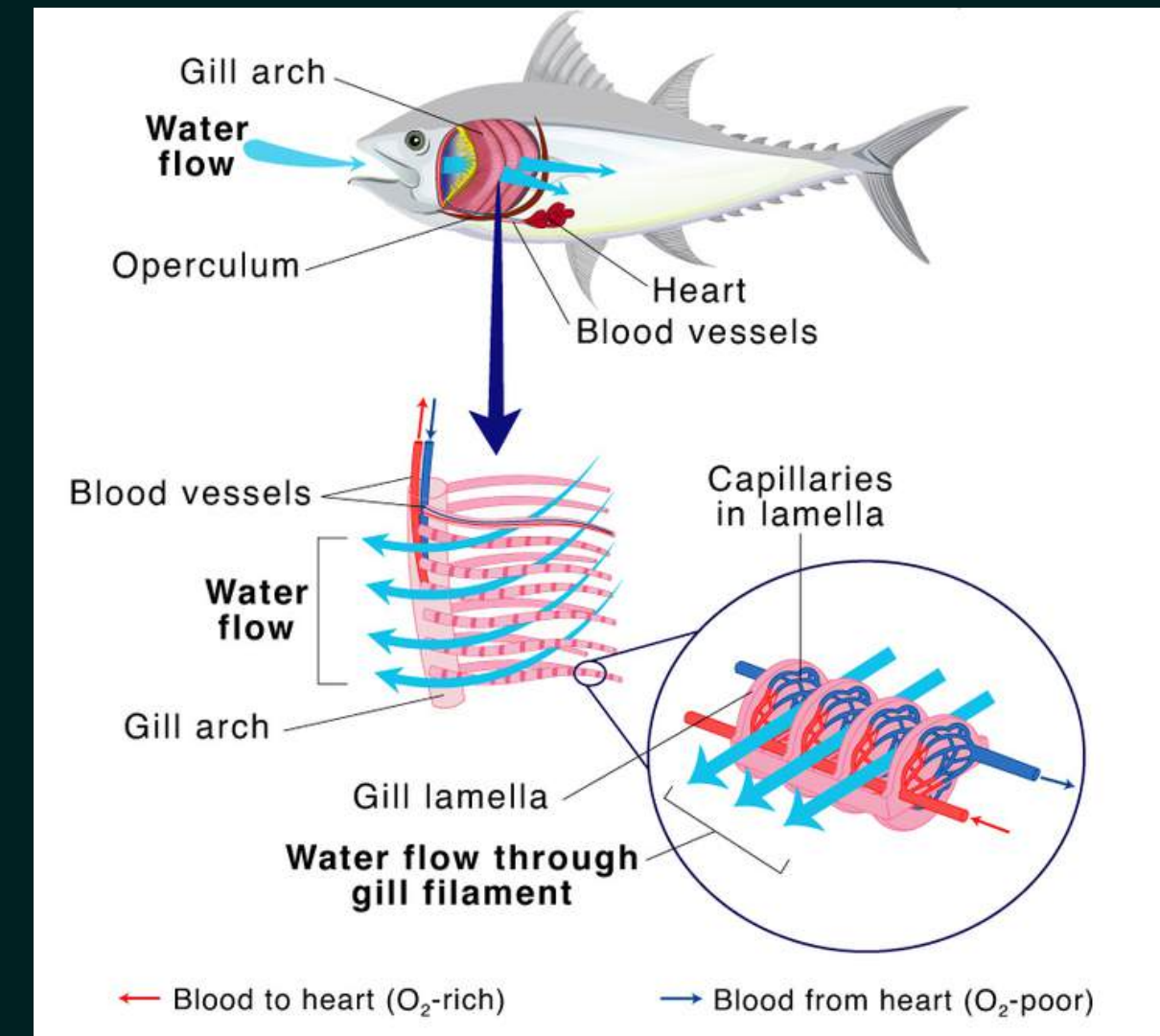
Q. Why is waterlogging harmful for plants?

Answer:

Waterlogging is harmful for plants because water fills the air spaces present between soil particles. Due to this, roots do not get enough oxygen for respiration. Without proper respiration, roots cannot release enough energy, so the plant may stop growing and can even die.

How do fishes breathe underwater?

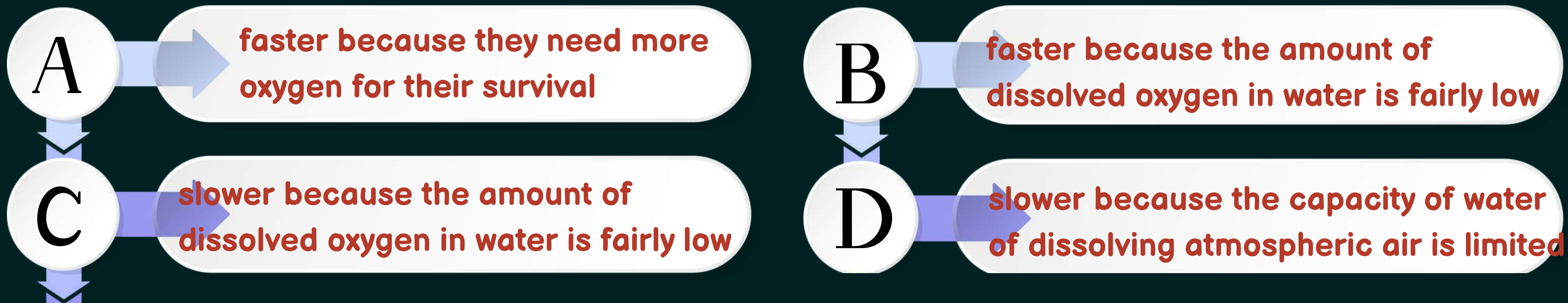
- Fishes are aquatic organisms.
- They use oxygen dissolved in water for respiration.
- Fishes take in water through their mouth.
- This water is forced to pass over the **gills**.
- At the gills, the dissolved oxygen present in water is taken up by the blood.
- Carbon dioxide produced in the body is removed through the same exchange process.
- The **gill-slits or operculum** behind the eyes open and close in coordination with the mouth to help water pass over the gills.
- **Since the amount of dissolved oxygen in water is fairly low compared to oxygen in air, the rate of breathing in aquatic organisms is much faster than in terrestrial organisms.**



LET'S TEST YOUR MEMORY



Q. As compared to terrestrial organisms, the rate of breathing in aquatic organisms is:



LET'S TEST YOUR MEMORY



Q. What advantage over an aquatic organism does a terrestrial organism have with regard to obtaining oxygen for respiration? [NCERT]

Answer:

Oxygen is more abundant in the air than in water.

Terrestrial organisms get oxygen more easily due to better respiratory surfaces (lungs, skin, etc.).

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Q. Rate of breathing in aquatic organisms is much faster than that in terrestrial organisms. Give reasons.

The rate of breathing in aquatic organisms is much faster because the amount of dissolved oxygen in water is much less than the amount of oxygen present in air. So, aquatic organisms have to breathe faster to take in enough oxygen for respiration.

LET'S TEST YOUR MEMORY



Q. Aerobic respiration requires intake of oxygen to break down food and release energy. In plants and human beings, exchange of gases takes place through special structures. In humans, breathing helps air reach the lungs, where exchange of gases takes place. The diaphragm helps in changing the size of the chest cavity. In plants, exchange of gases mainly occurs by diffusion. However, in large animals like humans, diffusion alone is not sufficient to transport oxygen to all parts of the body. Therefore, oxygen is transported with the help of blood and haemoglobin.

Q1. Name the structures through which gaseous exchange takes place in:

(a) Plants

(b) Human beings

Q2. Name the structure that controls the size of the chest cavity in humans.

Q3. What is the process by which gas exchange occurs in plants?

Q4. Why is diffusion alone not sufficient to carry oxygen throughout the human body?

Q5. How is oxygen transported to all parts of the body in humans?

Kal Padhenge
Dil ke baare
me❤️

